NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:29-12-24**
Time: 3HRS **GTM-5** **Max. Marks: 300**

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
D	B	A	D	C	C	B	D	D	A
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	C	A	C	A	A
21	22	23	24	25					
2	8	1	13	180					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	C	C	A	D	D	B	C	C	A
36	37	38	39	40	41	42	43	44	45
B	B	C	A	A	B	C	B	A	D
46	47	48	49	50					
3	8	3	4	7					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	C	C	B	C	C	B	B	C	D
61	62	63	64	65	66	67	68	69	70
A	C	B	C	B	C	D	B	A	B
71	72	73	74	75					
306	3	5	9	17					

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SOLUTIONS

PHYSICS

1 Magnetic moment M = NIA

$$M = I_0 \frac{\sqrt{3}}{4}\left(\sqrt{2}a\right)^2 = \frac{\sqrt{3}}{2} I_0 a^2$$

2 $H^2 + H^2 \rightarrow He^4 + E$

$$Q = \left(2m_H - m_{He}\right)c^2 = 23.83\ MeV$$

For one gram of He

$$E = \frac{1}{4} \times 6.022 \times 10^{23} \times 23.83\ MeV$$

Mass of coal

$$m = \frac{E}{30KJ\ /\ gm} \times 2 \times 10^4 kg$$

3 $S_{av} = \frac{P}{4\pi R^2} = \frac{1}{2} \in_0 cE_0^2$

$$\therefore E_0 = \sqrt{\frac{P}{2\pi R^2 \in_0 c}}$$

$$= \sqrt{\frac{3}{2 \times 3.14 \times 100 \times 8.85 \times 10^{-12} \times 3 \times 10^8}}$$

= 1.34 V/m

4 Mass of the sphere is given by $M = \frac{4}{3}\pi y^3 \rho$

Gravitational force $F = \frac{G\left(\frac{4}{3}\pi y^3 \rho\right)m}{y^2}$

$$a = \frac{F}{m}$$

5 $\frac{a_0}{g} = \tan\theta$

$$\Rightarrow a_0 = \frac{g}{\sqrt{3}}$$

6 Conceptual

7 Since ML = Pt

$$\Rightarrow L = \frac{Pt}{M}$$

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8 When one e^- is removed from neutral helium atom, it

For one e^- species we know

$$E_n = \frac{-13.6Z^2}{n^2} eV / atom$$

For helium ion, Z=2 and for first orbit n=1

$$\therefore E_1 = \frac{-13.6}{(1)^2} \times 2^2 = -54.4eV$$

$\therefore$ Energy required to removed this $e^- = +54.4eV$

$\therefore$ Total energy required = 54.4 + 24.6 = 79 eV

9 $$n = \frac{10^{-3}}{200 \times 10^6 \times 1.6 \times 10^{-19}}$$

10 use wheatstone bridge condition

11 $$v\cos\left(30^o\right) = v_0 \cos 60^o$$

$$v = \frac{v_0}{\sqrt{3}}$$

$$e = \frac{\frac{v_0}{2\sqrt{3}}}{\frac{v_0\sqrt{3}}{2}} = \frac{1}{3}$$

12 $$P = 2P_l + P_m$$

13 Conceptual

14 Applying conservation of energy

$$Mgh = \frac{1}{2}Mv_0^2 + \frac{1}{2}I\omega_0^2 = \frac{1}{2}M\left(\frac{5v_0}{4}\right)^2$$

$$\Rightarrow I = \frac{9MR^2}{16}$$

$$Mx^2 = I \Rightarrow x = \frac{3R}{4}$$

n = 3

15 For conservative forces, dU = - dW

$$U_f - U_i = -W_{i-f}$$

$$or\ W_{i\to f} = U_i - U_f = q(V_i - V_f)$$

$$5 \times 10^{-5} = 2 \times 10^{-6}\left[2a(0.1)^2 - 0\right]$$

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$or\ \ a = 1.25 \times 10^3 V/m^2 = 1250\ V/m^2$

16 $v\cos 37^0 = 20\cos 53^o$

$v\cos 37^o = 20 \times \frac{3}{5}$

$\Rightarrow v = 15 m/s$

17 $\rho = \frac{m}{\ell \pi r^2}$

$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta \ell}{\ell}$

$= \frac{0.003}{0.3} + \frac{2 \times 0.005}{0.5} + \frac{0.06}{6} = \frac{4}{100} = 4\%$

18 $\left[(n-1)t\right]\frac{D}{d} = \frac{\lambda(2D)}{d}$

$\Rightarrow \lambda = \frac{(n-1)t}{2} = 5892\ \overset{o}{A}$

19 $(2v_0) \times e \geq v_0 \times \sqrt{3}$

$\Rightarrow e \geq \frac{\sqrt{3}}{2}$

20 $\lambda = \frac{kT}{\sqrt{2}\pi D^2 p}$

$\Rightarrow \lambda p = \text{constant}$

21 $I_1 = \frac{20}{10} A$

22 Wave velocity as a function of distance (x) from top is

$v = \sqrt{\frac{T}{\mu}} = \sqrt{g(9-x)}$

$\therefore \int_0^h \frac{dx}{\sqrt{g(9-x)}} = \int_0^{\sqrt{\frac{2h}{g}}} dt \left(\text{time taken by stone} = \sqrt{\frac{2h}{g}} \right)$

$\Rightarrow h = 8m$

23 $\vec{B} = \vec{B}_1 + \vec{B}_2$

$\vec{B} = 0 \Rightarrow \vec{B}_1 = -\vec{B}_2$

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$$\frac{\mu_0 I_0}{2\pi d}+\frac{\mu_0 I}{2R}=0$$

$$I_1=\frac{-I_0 R}{\pi d}$$

24 Heat is extracted from the source in path DA and AB is

$$\Delta Q=\frac{3}{2}R\left(\frac{P_0V_0}{R}\right)+\frac{5}{2}R\left(\frac{2P_0V_0}{R}\right)=\frac{13}{2}P_0V_0$$

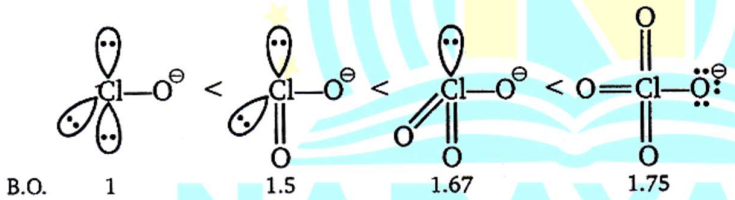
25 $s_1=\frac{1}{2}(60)(18)$

$s_2=40\times18$

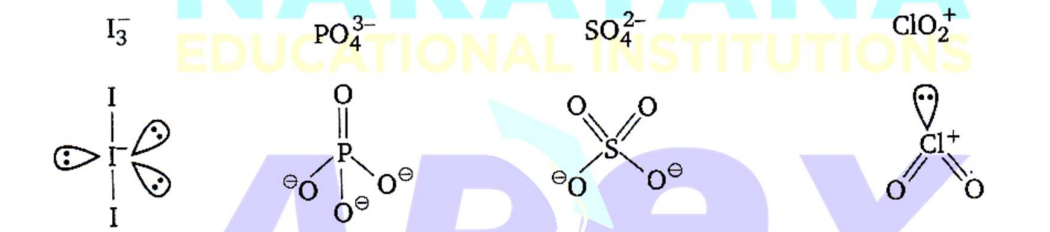
CHEMISTRY

26 Hydrated size of ion $\propto$ Charge density of ion. $K^+(aq)<Na^+(aq)<Li^+(aq)$

27



28



29 $\frac{S_2}{S_1}=\frac{p_2}{p_1}$

$\therefore S_2=S_1\frac{p_2}{p_1}$

$S_2=5.3\times10^{-4}\times\frac{1.00}{0.75}=7.07\times10^{-4}M$

30 In the reaction at cathode of Leclanche cell, manganese is reduced from +4 oxidation state to +3 state.

31 Increasing order of crystal field splitting energy is : $H_2O < NH_3 <$ en

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Thus, increasing order of energy for the given complexes is :

$\left[Co\left(H_2O\right)_6\right]^{3+} < \left[Co(NH_3)_6\right]^{3+} < \left[Co(en)_3\right]^{3+}$

As, $E = \frac{hc}{\lambda}$

Thus, increasing order of wavelength of absorption is :

$\left[Co(en)_3\right]^{3+} < \left[Co(NH_3)_6\right]^{3+} < \left[Co\left(H_2O\right)_6\right]^{3+}$

32 Read in optical isomerism of type $\left[M\left(AA\right)_2 B_2\right]$ and $\left[M\left(AA\right)_3\right]$. Here AA is a symmetrical didentate ligand such as 'en'.

33 Third period contains 3s and 3p subshell. Total 4 orbitals and s elements in third period. Fourth period contains 4s, 3d and 4p subshell. Total 9 orbitals and 18 elements in fourth period.

34 No. of stereocenters = 5
No. of stereoisomers = $2^n = 2^5$

35 Conceptual

36 Conceptual

37 The dehydrohalogenation of 1 Bromo propane with alc. NaOH or KOH gives propene.
Hydrohalogenation with HBr gives 2-bromopropane due to addition of Markovnikov's rule of addition.

38 Adiabatic irreversible expansion and compression of an ideal gas

39 H^+ ions provide acidic medium to the reactant.

40 $Pb_3O_4 + 8HCl \rightarrow 3PbCl_2 + Cl_2 + H_2O$

41 When 2 molecules of the same or different aldehyde or ketone condense in the presence of dil. Alkali, it is called aldol condensation.
When the same molecules of aldehyde or ketone condense in the presence of dil. Alkali, it is called aldol or ketol when ketones are involved.
Products are formed which have both aldehyde or ketone and alcohol.
They lose water at high temperature to give α,β-unsaturated carbonyl compounds.
The reaction will have (b) as a product as it has the ketone at α-carbon and the unsaturation at the β-position in the condensation of intramolecular ketone.

42 When aniline is treated with $NaNO_2$ in the presence of acids then benzene diazonium salts are formed which react with other compounds like phenols, or para-methyl phenols in the presence of alkali, it results in a coupling reaction that produces colored dyes called azodyes. Here the diazonium salt reacts with paramethyl phenol to produce an azodye. The nitrogen gets attached at the ortho

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position as the methyl group activates that position.

43 Conceptual

44 $\Delta H = \left(\sum BE\right)_R - \left(\sum BE\right)_P$

$10 = \left(\frac{1}{2}\times 104 + \frac{1}{2}\times 118\right)_R - (O-H)_P$

O – H = 52 + 59 – 10 = 101 kcal/mol

45 $P_1 = \frac{nRT}{V_1} = \frac{2\times 0.082\times 243.6}{20} = 2\ atm$

$\therefore \Delta S = 2.303\ nR\log\frac{P_1}{P_2}$

$\Delta S = 2.303\times 2\times 2\times \log\frac{2}{1}$

$= 2.303\times 4\times 0.3010$

$\Delta S = 2.77\ cal\ /\ K$

46 Compounds undergoes various chemical or physical at room temperature are said to be unstable at room temperature. So anti aromatic compounds unstable at room temperature.

47 x = 3, y = 2, z = 3

Hence x + y + z = 3 + 2 +3 = 8

48 Acidic buffer

$pH = pK_a + \log\frac{[Salt]}{[Acid]}$

$= 3.7 + \log\frac{0.2}{1} = 3.7 + \log\frac{1}{5}$

= 3.7 – log 5 = 3

49 Let, dilution is done x-times

$\pi = \frac{n_2}{V}ST\left(n_2 = solute = \text{constant}\right)$

$\pi_1 = \frac{n_2}{V}\times S\times T_1$

$\pi_2 = \frac{n_2 ST_2}{xV}$

$\frac{\pi_1}{\pi_2} = x\frac{T_1}{T_2}$

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$$\frac{400}{105.3} = x\left(\frac{293}{308}\right)$$

$\therefore \quad x = 4$

Thus, four times dilution.

50 Seven $\left({}_{64}Gd = [Xe]4f^7\,6s^2\,5d^1\right)$

MATHEMATICS

51 $$I = \int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx$$

Using King and add

$$2I = \int_{-\pi}^{\pi} \frac{2x(1+\sin x)+2(-x)(1-\sin x)}{1+\cos^2 x}dx = 4\int_{-\pi}^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$$

$$I = 2\int_{-\pi}^{\pi} f(x)dx$$

$$I = 4\int_{0}^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$$

Using Kind and add

$$2I = 4\pi\int_{0}^{\pi} \frac{\sin x}{1+\cos^2 x}dx$$

Put cos x = t

$$2I = 4\pi\int_{-1}^{1} \frac{dt}{1+t^2}$$

$$\therefore \; I = 4\pi \times \frac{\pi}{4} = \pi^2$$

52 $$T_1 = \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) = \tan^{-1}\left(\frac{1}{3}\right) = \tan^{-1}2 - \tan^{-1}1$$

$$T_2 = \sin^{-1}\left(\frac{1}{\sqrt{50}}\right) = \tan^{-1}\left(\frac{1}{7}\right) = \tan^{-1}3 - \tan^{-1}2$$

$$T_{10} = \tan^{-1}101 - \tan^{-1}100$$

$$Sum = \tan^{-1}101 - \tan^{-1}1 = \tan^{-1}\left(\frac{100}{102}\right) = \tan^{-1}\left(\frac{50}{51}\right)$$

$\therefore \quad p + q = 50 + 51 = 101$

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53 $y=\left(x+\sqrt{1+x^2}\right)^n$

$$\frac{dy}{dx}=n\left(x+\sqrt{1+x^2}\right)^{n-1}\left(1+\frac{x}{\sqrt{1+x^2}}\right)=\frac{n\left(x+\sqrt{1+x^2}\right)^n}{\sqrt{1+x^2}}$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{ny}{\sqrt{1+x^2}};\ \left(1+x^2\right)\frac{dy}{dx}=ny\sqrt{1+x^2}$$

$$\left(1+x^2\right)\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=n\sqrt{1+x^2}+\frac{nxy}{\sqrt{1+x^2}}=n\sqrt{1+x^2}\,\frac{ny}{\sqrt{1+x^2}}+x\frac{dy}{dx}$$

$$\Rightarrow \quad \left(1+x^2\right)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=n^2y$$

54 $e^x=\dfrac{f'(x)}{1-f'(x)};\ f'(x)=\dfrac{e^x}{1+e^x};$

$f(x)=\ln(1+e^x)+C$

$f(0)=0 \Rightarrow \ln(2)+C=0 \quad \Rightarrow C=-\ln 2$

$f(x)=\ln(1+e^x)-\ln 2 \Rightarrow \lim\limits_{x\to 0}(1+f(x))^{1/x}=\sqrt{e}$

55 $$I=\int \frac{3(\tan x-1)\sec^2 x}{(\tan x+1)\sqrt{\tan^3 x+\tan^2 x+\tan x}}dx=3\int\frac{(t-1)}{(t+1)\sqrt{t^3+t^2+t}}dt$$

$$=3\int \frac{\left(1-\frac{1}{t^2}\right)}{\left(t+\frac{1}{t}+2\right)\sqrt{t+\frac{1}{t}+1}}dt$$

Let $t+\dfrac{1}{t}+1=z^2 \Rightarrow \left(1-\dfrac{1}{t^2}\right)dt=2zdz$

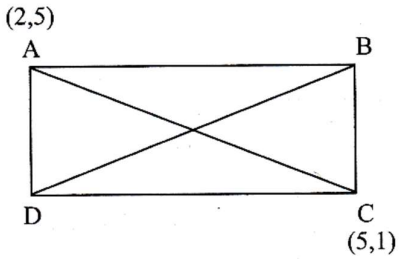
$$=6\int\frac{dz}{\left(z^2+1\right)}=6\tan^{-1}\sqrt{t+\frac{1}{t}+1}+C$$

56 $\because$ diagonal bisect each other

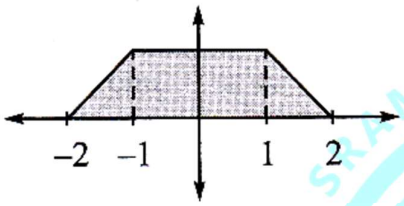
$\therefore$ mid-point on AC i.e., $\left(\dfrac{7}{2},3\right)$ will be on y = 2x + k

$\therefore$ $3 = 7 + k \Rightarrow k = -4$

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57



Area bounded $= \frac{1}{2}\times 1\times 1 + 2\times 1 + \frac{1}{2}\times 1\times 1 = 3$

58 If $\sin\theta$ *and* $\cos\theta$ are roots of $3x^2 - x + k = 0$, then

$\sin\theta + \cos\theta = \frac{1}{3}$ …(1)

$\sin\theta\cos\theta = \frac{k}{3}$ …(2)

$(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$

$\frac{1}{9} = 1 + \frac{2k}{3} \Rightarrow k = \frac{-4}{3}$

Also $\sin^3\theta + \cos^3\theta = (\sin\theta + \cos\theta)(\sin^2\theta + \cos^2\theta - \sin\theta\cos\theta)$

$= \frac{1}{3}\left(1 + \frac{4}{9}\right) = \frac{13}{27}$

$\Rightarrow \quad 54(\sin^3\theta + \cos^3\theta) = 26$

59 $\vec{a} = \vec{b}\times\vec{c} + 2\vec{b}$

Taking dot product with $\vec{b}$

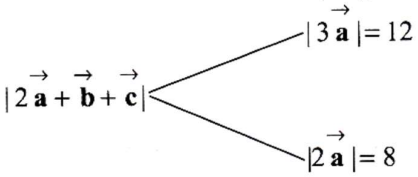
$\vec{a}\cdot\vec{b} = 2|\vec{b}|^2 \Rightarrow |\vec{a}||\vec{b}|\cos\theta = 2|\vec{b}|^2$

$\Rightarrow \quad \cos\theta = \frac{4}{|\vec{a}|} \Rightarrow |\vec{a}| = 4 \Rightarrow \theta = 0^o$

$\Rightarrow \quad \vec{a} = 2\vec{b}$

Now, $\vec{b}\times\vec{c} = 0 \Rightarrow \vec{b} = \vec{c}$ *or* $\vec{b} = -\vec{c}$

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$\therefore$ Required sum = 12 + 8 = 20

60 Favourable cases are (2, 3), (2, 6), (4, 3), (4, 6), (6, 3), (6, 6) and (3, 2), (6, 2), (3, 4), (6, 4), (3, 6).

$\therefore$ Probability $=\frac{11}{36}$

61 $\because \ |z+16|^2=16|z+1|^2$

$\Rightarrow \quad z\bar{z}+16z+15\bar{z}+256=16\left(z\bar{z}+z+\bar{z}+1\right)$

$\Rightarrow \quad 15z\bar{z}=240 \ \Rightarrow \ z\bar{z}=16 \ \Rightarrow \ |z|=4$

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62. $ABA + AB^2 = 3(A + B)$

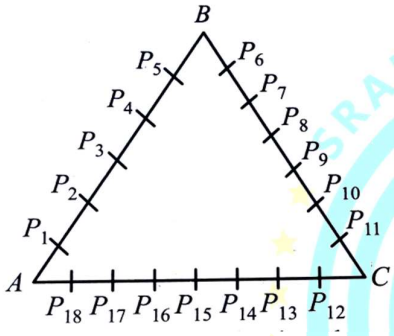
$AB(A + B) - 3(A + B) = 0$

$(AB - 3I)(A + B) = 0$

If $|A + B| \neq 0$ then $AB = 3I$

$\Rightarrow A, B$ are non-singular and commutative.

63.



Number of ways selecting three points from side $AB = {}^5C_3$

Number of ways selecting three points from side $BC = {}^6C_3$

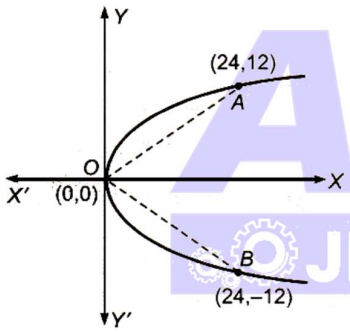
Number of ways selecting three points from side $AC = {}^7C_3$

Total number of triangle possible formed using the points $P_1P_2 \ldots P_{18}$

$= {}^{18}C_3 - {}^5C_3 - {}^6C_3 - {}^7C_3$

$= 816 - 10 - 20 - 35 = 751$

64. $y^2 = 6x$ has origin as its vertex.



Points with abscissa = 24

$y^2 = 6 \times 24 = 6^2 \times 2^2$

$y = \pm 12$

Slope of OA $= \dfrac{12 - 0}{24 - 0} = \dfrac{1}{2}$

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Slope of OB $= \dfrac{-12-0}{24-0} = -\dfrac{1}{2}$

$y = \pm\dfrac{1}{2}x$

65. $[x] - \left[\dfrac{x}{3}\right]$ is discontinuous at $x = \{1,2,3,...,24\} - \{3,6,9,...,24\}$

$\left\{\dfrac{x}{11}\right\} + 14\left[\dfrac{x}{13}\right]$ is discontinuous at $x = \{11,22,13\}$

At $x = 11$; $RHL = (0+0)(11-3) = 0$

$LHL = (1+0)(10-3) = 7$

At $x = 13$; $RHL = \left(\dfrac{2}{11}+14\right)(13-4) = \dfrac{156\times 9}{11}$

$LHL = \left(\dfrac{2}{11}+0\right)(12-4) = \dfrac{16}{11}$

At $x = 22$; $RHL = (0+14)(22-7) = 14\times 15$

$LHL = (1+14)(21-7) = 15\times 14$

$\therefore$ Continuous at $x = 22$

$\therefore$ No. of points of discontinuous =15

66. $x = \dfrac{1}{9}[999.....99] = \dfrac{1}{9}[10^{20}-1]$ (i)

$y = \dfrac{1}{3}[999.....99] = \dfrac{1}{3}[10^{10}-1]$

$z = \dfrac{2}{9}[10^{10}-1]$

$\therefore \dfrac{x-y^2}{z} = \dfrac{(10^{20}-1)-(10^{10}-1)^2}{2(10^{10}-1)}$

$= \dfrac{(10^{10}+1)-(10^{10}-1)}{2} = 1$

67. Let the equation of the circle whose equation is to be find out be

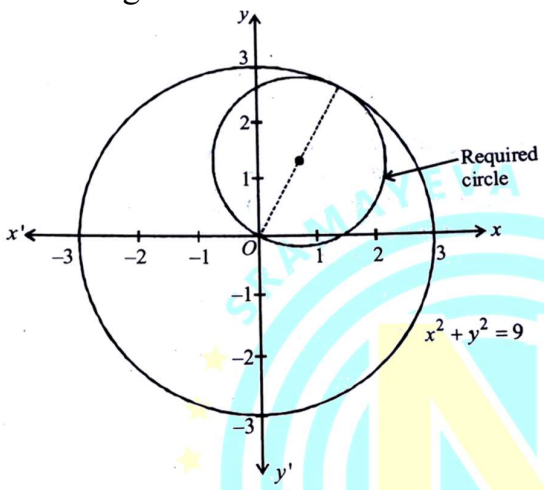
$x^2 + y^2 + 2gx + 2fy + c = 0$

As this circle passes through (0, 0) and (1, 0),

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$\therefore\ c=0,\ g=-\dfrac{1}{2}$

Since the required circle touches the given circle $x^2+y^2=9$ internally like as shown in the figure.



$\therefore\ 2\times$ radius of required circle = radius of given circle

$\Rightarrow \quad 2\sqrt{g^2+f^2}=3\Rightarrow g^2+f^2=\dfrac{9}{4}$

68. Given set A with equation $|x+1|<2$ and set B with equation $|x-1|\geq 2$.

$A: x\in(-3,1)$

$B: x\in(-\infty,-1]\cup[3,\infty)$

Take, $B-A=(-\infty,-3]\cup[3,\infty)=R-(-3,3)$

69. Put y = 0 in $\dfrac{x}{7}+\dfrac{y}{2\sqrt{6}}=1$, we get x = 7;

So, ellipse intersect x-axis at 7, i.e., a = 7

Put x = 0 in $\dfrac{x}{7}-\dfrac{y}{2\sqrt{6}}=1$, we get

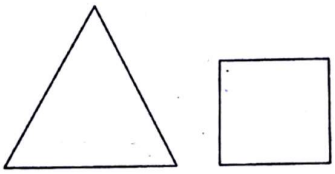
$y=-2\sqrt{6}$ so, ellipse intersect y-axis at $-2\sqrt{6}$ i.e., $b=-2\sqrt{6}$

$\therefore\ e=\sqrt{1-\dfrac{b^2}{a^2}}=\sqrt{1-\dfrac{24}{49}}=\dfrac{5}{7}$

70. Since, length of the given wire = 22 m

Let xm length is used in equilateral Δ and rest for the square

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Let side of an equilateral $\Delta = a$ and side of a square = b, $\Rightarrow$ 3a = x & 4b = 22 – x

Now, total area $= \frac{\sqrt{3}}{4}a^2 + b^2$

$= \frac{\sqrt{3}}{4}x^2/9 + \frac{(22-x)^2}{16}$

$\frac{dA}{dx} = 0 \Rightarrow x\left(\frac{\sqrt{3}}{2\times 9} + \frac{1}{8}\right) = \frac{22}{8} \Rightarrow x\left(\frac{4\sqrt{3}+9}{36}\right) = \frac{11}{2}$

a = x/3

$a = \left(\frac{11/2}{\frac{4\sqrt{3}+9}{36}}\right)\left(\frac{1}{3}\right) = \frac{66}{4\sqrt{3}+9}$

71. Given the shortest distance = 13

$$\Rightarrow \frac{\left\|\begin{matrix} 0 & 4 & 1 \\ 3 & -4 & 0 \\ 2\lambda & 3 & -12 \end{matrix}\right\|}{\left\|\begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 4 & 1 \\ 3 & -4 & 0 \end{matrix}\right\|} = 13$$

$$\Rightarrow \frac{|153+8\lambda|}{|4\hat{i}+3\hat{j}-12\hat{k}|} = 13 \Rightarrow \frac{|153+8\lambda|}{13} = 13 \Rightarrow |153+8\lambda| = 169$$

$$\Rightarrow 153+8\lambda = 169, -169 \Rightarrow \lambda = \frac{16}{8}, \frac{-322}{8}$$

So, $8\left|\sum_{\lambda\in S}\lambda\right| = 306$

72. $x^2dy + y^2dx = 0$

$\Rightarrow \quad \frac{dy}{y^2} + \frac{dx}{x^2} = 0$

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$\Rightarrow \quad \frac{-1}{y}-\frac{1}{x}=C$

$(2,2) \Rightarrow C=1$

$\therefore \quad \frac{1}{y}+\frac{1}{x}=1$

$\Rightarrow \quad y=\frac{x}{x-1}$

Required area $=\int_2^3 \frac{x}{x-1}dx=\int_2^3\left(1+\frac{1}{x-1}\right)dx$

$\Rightarrow \left(x+\ln(x-1)\right)_2^3=1+\ln 2 \equiv a+\ln b$

Hence, a + b = 3

73. $T_{r+1}={}^6C_r\cdot\left(x^2\right)^{6-r}\cdot\left(\frac{-3}{x}\right)^r$

$12 - 3r = 0$

$\Rightarrow \quad r = 4$

$T_5={}^6C_4\cdot(-3)^4 \Rightarrow 5\cdot 3^5$

74. $f(x)=\sec^2 x+4\text{cosec}^2 x=1+\tan^2 x+4\left(1+\cot^2 x\right)$

$=5+\tan^2 x+4\cot^2 x$

Hence $f(x)_{\min}=5+4=9$

75. Given variance and mean are 3 & 8

Variance $=\sigma^2=\frac{\sum_{r=1}^{15}x_r^2}{15}-(\bar{x})^2$

Put $\bar{x}=8$ *and* $\sigma^2=3$,

$\frac{\sum x_r^2}{15}-8^2=3^2 \Rightarrow \sum x_r^2=1095$

Now, the new $\sum x_r^2=1095-5^2+20^2=1470$

And, new $\sum x_r=(15\times 8)-5+(20)=135$

$\therefore$ Variance $=\frac{1470}{15}-\left(\frac{135}{15}\right)^2=98-81=17$

Therefore, the variance is 17.

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